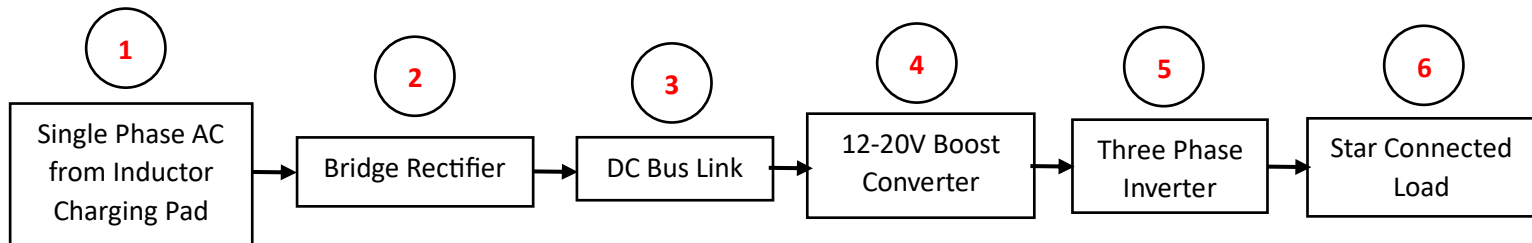


Full Description and Explanation of the Variable Frequency Drive



Stage 1→2: A typical diode bridge using Schottky Diodes to convert single phase AC to DC.

Stage 2→3: DC ripple was reduced by placing a capacitor in parallel, the charging and discharging of the capacitor smoothens out the voltage.

Stage 4: The voltage in from the inductor pads is 12V but we require 20V at the drain of the High side N-MOS. This requires a voltage boost converter utilising an LTC3872 Chip in which the boost is determined by a voltage divider.

Stages 1→4: Are primarily for adapting and stabilising the voltage input from the charging pads.

Stage 5→6: This is where the V/F control is incorporated for the three-phase YASA PMSM motor.

Stage 5: A 6 N-MOS arrangement is used to control the three phases of the AC output. We use a signal generator (ESP-32) to generate reference sinusoidal voltage signals which have the required frequency for the motor and are each phase shifted by 120 degrees. A high frequency triangle wave is formed (at 20Khz) and comparisons are made between each reference voltage and the triangle wave. At any time, the reference voltage is greater than the triangle waveform, the High side N-MOS activates. Any time it's lower than the triangle waveform, the low side N-MOS activates. A small dead time is included in the comparison to prevent both the High side and Low side from conducting at the same time (this is to prevent a short).

To control the V/F output of the waveform, we reduce/increase the frequency of the reference voltages and reduce/increase the ratio of the voltage between the reference and the triangle wave.

Snubber RC circuits are placed in parallel of each N-MOS to block out voltage transients and smoothen the AC output. Resistors are placed in series on each gate of the N-MOS to reduce the current in.

Stage 6: The output is modelled to a three-phase star connected load that has a small inductive and resistive load.

Additional Information:

Gate driver selected: IR2110, each one requires two GPIO's from the ESP-32 to control HO and LO.

N-MOS Selected: IRFZ44N, VDD = 55V, rated for the power requirement of the motor.